Linear Transformation

Let, U and V be two vector spaces over the same field F. A linear Transformation T of U into V, written as $T: U \to V$, is a function of U into V such that:

(i) $T(u_1+u_2) = T(u_1)+T(u_2)$ for all $u_1, u_2 \in V$.

(ii) $T(\alpha u) = \alpha T(u)$ for all $u \in U$ and for all $\alpha \in F$

Q. Let, $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the function (transformation) defined by:

$T(x,y,z) = (x+y-z, 0, x+2y-z)$.

Show that T is a linear transformation.

Soln. Let, $u_1, u_2 \in \mathbb{R}^3$ where $u_1 = (x_1, y_1, z_1)$ and $u_2 = (x_2, y_2, z_2)$

$u_1+u_2 = (x_1+x_2, y_1+y_2, z_1+z_2)$.

$T(u_1+u_2) = T(x_1+x_2, y_1+y_2, z_1+z_2)$

$= (x_1+x_2+y_1+y_2-z_1-z_2, 0, x_1+x_2+2y_1+2y_2-z_1-z_2)$

$T(u_1)+T(u_2) = T(x_1, y_1, z_1) + T(x_2, y_2, z_2)$

$= (x_1+y_1-z_1, 0, x_1+2y_1-z_1) + (x_2+y_2-z_2, 0, x_2+2y_2-z_2)$

$= (x_1+x_2+y_1+y_2-z_1-z_2, 0, x_1+x_2+2y_1+2y_2-z_1-z_2)$

Now, we have to show : $T(\alpha u_1) = \alpha T(u_1)$

$\alpha u_1 = \alpha(\alpha x_1, \alpha y_1, \alpha z_1)$

$T(\alpha x_1, \alpha y_1, \alpha z_1)$

$= (\alpha x_1 + \alpha y_1 - \alpha z_1, 0, \alpha x_1 + 2\alpha y_1 - \alpha z_1)$

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$R.S. = \alpha T(u_1)$

$= \alpha T(x_1, y_1, z_1)$

$= (\alpha x_1 + \alpha y_1 - \alpha z_1, 0, \alpha x_1 + 2\alpha y_1 - \alpha z_1)$

$\therefore T(u_1 + u_2) = T(u_1) + T(u_2)$

$\therefore T(\alpha u_1) = \alpha T(u_1)$ for all $u_1, u_2 \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$

$\therefore T$ is a linear transformation.

Q. Let $T: \mathbb{R}^2 \to \mathbb{R}^3$ be the function defined by:

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ x+y \\ x-y \end{pmatrix}$$

Show that: T is a linear transformation.

Soln: Let, $u_1, u_2 \in \mathbb{R}^2$ and $\alpha \in \mathbb{R}$.

when $u_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $u_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$

We have to show: $T(u_1 + u_2) = T(u_1) + T(u_2)$

$T(\alpha u_1) = \alpha T(u_1)$

$$u_1 + u_2 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

$$\therefore T(u_1 + u_2) = T\begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ x_1 + x_2 + y_1 + y_2 \\ x_1 + x_2 - y_1 - y_2 \end{pmatrix}$$

$$T(u_1) + T(u_2) = T\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + T\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$$

$$= \begin{pmatrix} x_1 \\ x_1 + y_1 \\ x_1 - y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ x_2 + y_2 \\ x_2 - y_2 \end{pmatrix}$$

$$= \begin{pmatrix} x_1 + x_2 \\ x_1 + x_2 + y_1 + y_2 \\ x_1 + x_2 - y_1 - y_2 \end{pmatrix}$$

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$\alpha u_1 = \begin{pmatrix} \alpha x_1 \\ \alpha y_1 \end{pmatrix}$

$T(\alpha u_1) = T\begin{pmatrix} \alpha x_1 \\ \alpha y_1 \end{pmatrix}$

$= \begin{pmatrix} \alpha x_1 \\ \alpha x_1 + \alpha y_1 \\ \alpha x_1 - \alpha y_1 \end{pmatrix}$

$\alpha T(u_1)$

$= \alpha T\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$

$= \alpha \begin{pmatrix} x_1 \\ x_1 + y_1 \\ x_1 - y_1 \end{pmatrix} = \begin{pmatrix} \alpha x_1 \\ \alpha x_1 + \alpha y_1 \\ \alpha x_1 - \alpha y_1 \end{pmatrix}$

$\therefore$ T is linear transformation.

Q. Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the function (transformation) defined by:

$T(x, y, z) = (x+1, x+y, x+z)$.

Show that T is not a linear transformation.

Soln: $u_1, u_2 \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$

$u_1 = (2, 1, 4)$; $u_2 = (1, 2, 3)$; $\alpha = 3$.

$T(\alpha u_1)$

$= T(2u_1)$

$= T(4, 2, 8)$

$= (5, 6, 12)$

$\alpha T(u_1) = 2 \cdot T(2, 1, 4)$

$= 2 \cdot (3, 3, 6)$

$= (6, 6, 12)$

So, T is not linear.

Q. Let, $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the function (transformation) defined by: $T(x, y, z) = (xy, x+y, x+z)$.

Show That: T is not a linear transformation.

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Linear Operator: Let, V be a vector spaces over a field F. If the transformation $T: V \to V$, is a linear transformation from V into itself, then T is called a linear operator.

Important:

A Linear Transformation be determined by the movement of Basis vectors:

Given, $T(1,0) = (-1,1)$ and $T(0,1) = (2,1)$, then find $T(x,y)$.

$\mathbb{R}^2 \rightarrow$ Basis $= \{(1,0), (0,1)\}$

$(x,y) = x(1,0) + y(0,1)$

$$\begin{aligned} T(x,y) &= T(x(1,0) + y(0,1)) \\ &= T(x(1,0)) + T(y(0,1)) \\ &= x\,T(1,0) + y\,T(0,1) \\ &= x(-1,1) + y(2,1) \end{aligned}$$

$\therefore T(x,y) = (-x+2y, x+y)$.

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Finding Kernel

*An important observation:

$$T(x,y) = (-x+2y, -2x+4y)$$

$T(0,0) = (0,0)$

$T(1,2) = (3,6)$

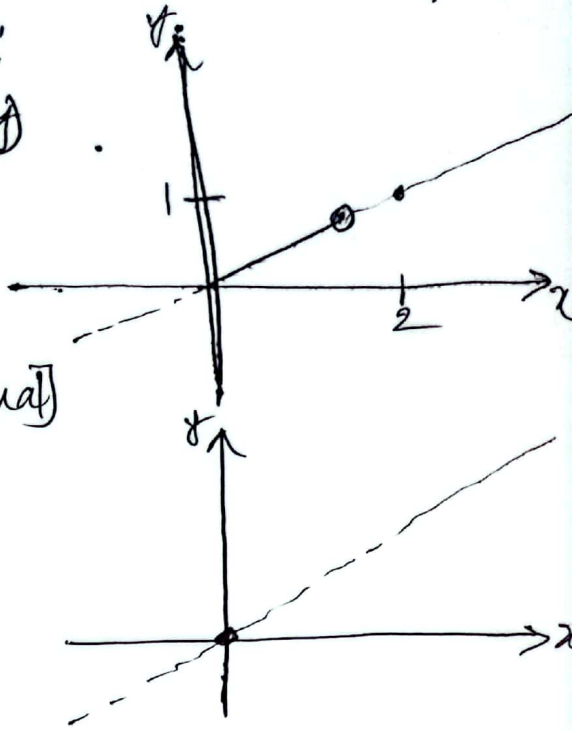
$\rightarrow T(2,1) = (0,0)$ [unusual]

multiple of (2,1): are Kernel

$T(6,3) = (0,0)$

$T(4,2) = (0,0)$

$T(1,\frac{1}{2}) = (0,0)$



* Formal Definition of Kernel of a Linear Transformation: Let, $T: V \rightarrow V$ be a linear transformation over the field F. Here, U is called the domain and V is called the codomain of T. Then the Kernel of T is defined by:

$$\text{Ker}(T) = \{u; u \in U \text{ and } T(u) = \vec{0}\}$$

* Matrix Representation of Linear Transformat

$T(x,y,z) = (x+y-z, x+2y, 2y-x+z)$.

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y-z \\ x+2y \\ -x+2y+z \end{pmatrix} = \begin{pmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$T \cong \begin{pmatrix} 1 & 1 & -1 \\ 1 & 2 & 0 \\ -1 & 2 & 1 \end{pmatrix}$$

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$$* T\begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} x+y-w \\ x+w \\ x-y+z \end{pmatrix}$$

$$= \begin{pmatrix} x+y & -w \\ x & +w \\ x-y+z & \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 & -1 \\ 1 & 0 & 0 & 1 \\ 1 & -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix}$$

Q. Let, $T: R^3 \to R^3$ be a linear transformation defined by $T(x,y,z) = (2x-y+z,\ x+2y-z,\ x+7y-4z)$

$$T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2x-y+z \\ x+2y-z \\ x+7y-4z \end{pmatrix} = \begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

For kernel $T\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0$

$$\Rightarrow \begin{pmatrix} 2 & -1 & 1 \\ 1 & 2 & -1 \\ 1 & 7 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 7 & -4 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 1 & 2 & -1 & 0 \\ 1 & 7 & -4 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} 2 & -1 & 1 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & 15 & -9 & 0 \end{array}\right)$$

$$\sim \left(\begin{array}{ccc|c} \boxed{2} & -1 & 1 & 0 \\ 0 & \boxed{5} & -3 & 0 \\ 0 & 15 & 0 & 0 \end{array}\right) \uparrow$$

$2x - y + z = 0$

$5y - 3z = 0$

$z = t$

$y = \frac{3t}{5}$

$2x = y - z$

$= \frac{3t}{5} - t$

$\Rightarrow 2x = \frac{-2t}{5}$

$\therefore x = \frac{-t}{5}$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = t \begin{pmatrix} -\frac{1}{5} \\ \frac{3}{5} \\ 1 \end{pmatrix}$$

$$= \text{span}\left\{ \begin{pmatrix} -\frac{1}{5} \\ \frac{3}{5} \\ 1 \end{pmatrix} \right\}$$

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· Kernel $(T) = \text{span}\left\{\begin{pmatrix} -1/5 \\ 3/5 \\ 1 \end{pmatrix}\right\}$

$\dim(\text{Kernel}(T)) = 1$

Nullity $= 1$

*One One Linear Transformation:

Let $T: U \to V$ be a linear transformation over the field F. Here, U is called the domain and V is called codomain of T. if

$\text{Ker}(T) = \vec{0}$ or $\text{Nullity}(T) = 0$.

then T is called One-One/Injective linear transformation.

*Finding Image/Range:

Definition: Let, $T: U \to V$ be a linear transformation over the field F. Here, U is called the domain and V is called the codomain of T. Then the image or Range is defined by:

$$\text{Im}(T) = \{T(u) : u \in V\}$$

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Q. Let: $T: R^3 \to R^3$ be a linear transformation defined by: $T(x,y,z) = (2x-y+z, x+2y-z, x+7y-4z)$
Find the Basis and Dimension of $Im(T)$.

Sol^n:

$$T\begin{pmatrix}x\\y\\z\end{pmatrix} = \begin{pmatrix}2x-y+z\\x+2y-z\\x+7y-4z\end{pmatrix} = \begin{pmatrix}2 & -1 & 1\\1 & 2 & -1\\1 & 7 & -4\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}$$

$$= \begin{pmatrix}2\\1\\1\end{pmatrix}x + \begin{pmatrix}-1\\2\\7\end{pmatrix}y + \begin{pmatrix}1\\-1\\-4\end{pmatrix}z$$

$$= \text{span}\left\{\begin{pmatrix}2\\1\\1\end{pmatrix}, \begin{pmatrix}-1\\2\\7\end{pmatrix}, \begin{pmatrix}1\\-1\\-4\end{pmatrix}\right\}$$

$Im(T)$ = column space of A
= Row space of A^T.

$$A^T = \begin{pmatrix}2 & 1 & 1\\-1 & 2 & 7\\1 & -1 & -4\end{pmatrix}$$

$$\sim \begin{pmatrix}2 & 1 & 1\\0 & 5 & 15\\0 & -3 & -9\end{pmatrix}$$

$$\sim \begin{pmatrix}2 & 1 & 1\\0 & 5 & 15\\0 & 0 & 0\end{pmatrix}$$

Row space of (A^T) = span$\{(2,1,1), (-1,2,7)\}$

Column space of A = span$\left\{\begin{pmatrix}2\\1\\1\end{pmatrix}, \begin{pmatrix}-1\\2\\7\end{pmatrix}\right\}$

Image(T) = span$\left\{\begin{pmatrix}2\\1\\1\end{pmatrix}, \begin{pmatrix}-1\\2\\7\end{pmatrix}\right\}$

Basis $= \left\{\begin{pmatrix}2\\1\\1\end{pmatrix}, \begin{pmatrix}-1\\2\\7\end{pmatrix}\right\}$

Dim of Image = Rank = 2.

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Q. Let $T: R^4 \to R^3$ be a linear transformation defined by:

$T(x,y,z,t) = (x-2y+2+3t, x+5y+3z-2t, 3x+y+5z+4t)$.

Find the Basis and Dimension of $Im(T)$.

* Onto Linear Transformation:

Let, $T: U \to V$ be a linear transformation over the field F. Here, U is called the domain and V is called codomain of T. If:

$Im(T) = V$ or $Rank(T) = Dim(V)$.

Then T is called Onto/Surjective Linear Transformation.

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Eigenvalue, Eigenvector

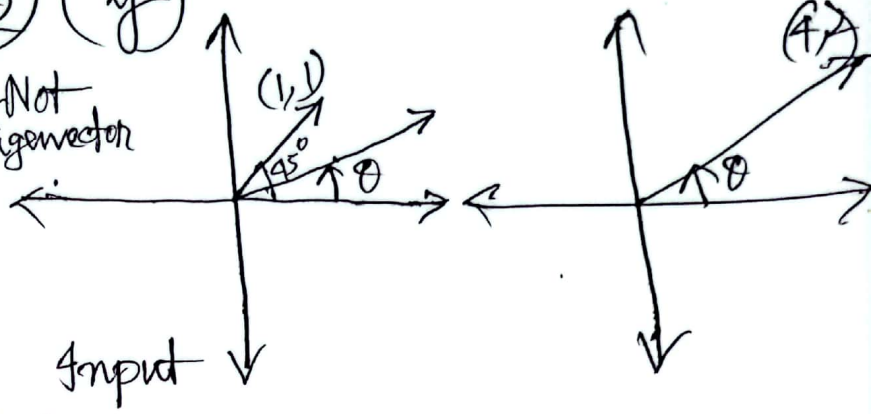
Consider, a matrix form:

$T(x,y) = (3x-2y, 2x-2y)$.

$$T\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}3 & -2\\ 2 & -2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$$

$$T\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}1\\0\end{pmatrix}$$ Not eigenvector

$$T\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}4\\2\end{pmatrix}$$



$$T\begin{pmatrix}2\\1\end{pmatrix} = \begin{pmatrix}4\\2\end{pmatrix} = \boxed{2}\begin{pmatrix}2\\1\end{pmatrix}$$

2 ← eigenvalue

eigenvector: A vector which does not change the direction.

$$T\begin{pmatrix}8\\4\end{pmatrix} = \begin{pmatrix}16\\8\end{pmatrix} = 2\begin{pmatrix}8\\4\end{pmatrix}$$

$$T\begin{pmatrix}3\\6\end{pmatrix} = \begin{pmatrix}-3\\-6\end{pmatrix} = (-1)\begin{pmatrix}3\\6\end{pmatrix}$$

→ Idea of Charecteristic Polynomial:

* Finding Eigenvalue:

Let, A be a square matrix of size $n \times n$.

If λ is an eigenvalue of A, then for any vector v of size $n \times 1$ then

$$Av = \lambda v \quad \Big| \quad Av - \lambda v = 0$$

$$\det(A - \lambda I) = 0 \quad \Big| \quad \Rightarrow (A - \lambda I)v = 0$$

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Q. Find the characteristic equation and eigenvalues of the matrix A.

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{pmatrix}$$

Now, $\det(A - \lambda I) = 0$

$$\Rightarrow \begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)(2-\lambda) - 2 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 4 = 0$$

$$\therefore \lambda = 4, 1.$$

Q. Find the characteristic equation and eigenvalues of the matrix A.

$$A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}$$

$$\therefore A - \lambda I = \begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix}$$

$$|A - \lambda I| = 0$$

$$\Rightarrow \begin{vmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)[(2-\lambda)(3-\lambda) - 2] - 1[0(3-\lambda) - 2(-1)] + 2[0(1) - (2-\lambda)(-1)] = 0$$

$\Rightarrow (1-\lambda)(\lambda^2 - 5\lambda + 6 - 2) - 1(0+2) + 2(2-\lambda) = 0$

$\Rightarrow (1-\lambda)(\lambda^2 - 5\lambda + 4) - 2 + 4 - 2\lambda = 0$

$\Rightarrow \lambda^2 - 5\lambda + 4 - \lambda^3 + 5\lambda^2 - 4\lambda - 2 + 4 - 2\lambda =$

$\Rightarrow -\lambda^3 + 6\lambda^2 - 11\lambda + 6 = 0$

$\Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\therefore \lambda = 1, 2, 3.$

$\therefore \lambda = 1, 2, 3.$

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Q. Find the eigenvectors of A where 1 and 4 are the eigenvalues of the matrix A. Also find the Basis of the eigenspaces of A.

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$$

Q. Find the characteristic equation and eigenvalues of the matrix A.

$$A = \begin{pmatrix} 2 & 2 & -5 \\ 3 & 7 & -15 \\ 1 & 2 & -4 \end{pmatrix}$$

Soln. $|A - \lambda I| = 0$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 2 & -5 \\ 3 & 7-\lambda & -15 \\ 1 & 2 & -4-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (2-\lambda)[(7-\lambda)(-4-\lambda) - (-15)2] - 2[3(-4-\lambda) - (-15)] + (-5)[(3\times 2) - (7-\lambda)] =$$

$$\Rightarrow (2-\lambda)[-28+4\lambda-7\lambda+\lambda^2+30] - 2[-12-3\lambda+15] - 5[6-7+7\lambda] = 0$$

$$\Rightarrow (2-\lambda)(\lambda^2-3\lambda+2) - 2(3-3\lambda) - 5(7\lambda-1) = 0$$

$$\Rightarrow 2\lambda^2 - 6\lambda + 4 - \lambda^3 + 3\lambda^2 - 2\lambda - 6 + 6\lambda - 35\lambda + 5 = 0$$

$$\Rightarrow -\lambda^3 + 5\lambda^2 - 37\lambda + 3 = 0$$

$$\Rightarrow \lambda^3 - 5\lambda^2 + 37\lambda - 3 = 0$$

. . . . . .

. . . . . .

. . . . . .

$\therefore \lambda =$ . . . . .

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*Properties of Eigenvalues:

Let, A be a square matrix and $\lambda_1, \lambda_2, \lambda_3, \ldots, \lambda_n$ be the eigenvalues of A, then:

$\text{Trace}(A) = \lambda_1 + \lambda_2 + \lambda_3 + \cdots + \lambda_n$

$\text{Det}(A) = \lambda_1 . \lambda_2 . \lambda_3 \ldots \ldots \lambda_n$

Q. Verify the properties of eigenvalues with $A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$ with $\lambda_1 = 1$ and $\lambda_2 = 4$.

Soln: Trace $= 3 + 2 = 5$ | $\lambda_1 + \lambda_2 = 5$

$\det = 4$ | $\lambda_1 \lambda_2 = 4$

Q. Verify the properties of eigenvalues with $A = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}$ and with $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$.

Soln: Trace $= 1 + 2 + 3 = 6$ | $\lambda_1 + \lambda_2 + \lambda_3 = 1 + 2 + 3 = 6$

$\det(A) = 6$ | $\lambda_1 \lambda_2 \lambda_3 = 1 . 2 . 3 = 6$.

Eigenvalues of a triangular matrix:

Let, A be an upper ~~aw~~ or lower triangular matrix: $A = \begin{pmatrix} \lambda_1 & * & * \\ 0 & \lambda_2 & * \\ 0 & 0 & \lambda_3 \end{pmatrix}$ or $A = \begin{pmatrix} \lambda_1 & 0 & 0 \\ * & \lambda_2 & 0 \\ * & * & \lambda_3 \end{pmatrix}$

Then the diagonal elements are the eigenvalues.

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Finding Eigenvectors:

Let, A be a square matrix of size $n \times n$. If λ is an eigenvalue of A, then the corresponding eigenvector v satisfies:

$Av = \lambda v \Rightarrow (A - \lambda I)v = 0$

That means v is a non-trivial solution of $(A - \lambda I)v = 0$

Q. Find the eigenvectors of A where 1 and 4 are the eigenvalues of matrix A. Also find the basis of eigenspace of A.

$$A = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix}$$

$$\therefore A - \lambda I = \begin{pmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{pmatrix}$$

Condition: $(A - \lambda I)v = 0$

For $\lambda = 1$, let $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be the eigenvector.

$$\begin{pmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \begin{matrix} 2x_1 + x_2 = 0 \\ x_2 = t \text{ [free variable]} \end{matrix}$$

$$\Rightarrow \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 2 & 1 & 0 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right)$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -t/2 \\ t \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} t/2 .$$

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For $\lambda=4$, let $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ be the eigenvector.

$$\begin{pmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{cc|c} -1 & 1 & 0 \\ 2 & -2 & 0 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right) \quad [R_2' = R_2 + 2R_1]$$

$\therefore -x_1 + x_2 = 0$

$x_2 = t$ [free variable]

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}=\begin{pmatrix} t \\ t \end{pmatrix}=\begin{pmatrix} 1 \\ 1 \end{pmatrix}t$$

Q. Find the eigenvectors of A where 1, 2, 3 are the eigenvalues of the matrix A. Also find the basis of eigenspace of A.

$$A=\begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix}$$

$$A-\lambda I=\begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix}$$

For $\lambda=1$, let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector.

$$\begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & 1 & 2 \\ 0 & 1 & 2 \\ -1 & 1 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ -1 & 1 & 2 & 0 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$-x_1 + x_2 + 2x_3 = 0$

$x_2 + 2x_3 = 0$

$x_3 = t$ [free variable]

$x_2 = -2t$

$x_1 = -2t + 2t = 0$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ -2t \\ t \end{pmatrix}=\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}t$$

$$V_1=\begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix}$$

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For $\lambda=2$, let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector

$$\begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -1 & 1 & 2 \\ 0 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right)$$

$$\Rightarrow \left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right) \quad [R_3' = R_3 - R_1]$$

$$-x_1 + x_2 + 2x_3 = 0$$
$$2x_3 = 0$$

$x_3 = t$ [free variable]

$x_3 = 0$

$x_1 = t$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} t \\ t \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} t$$

$$v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

For $\lambda=3$

$$\begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2 & 1 & 2 \\ 0 & -1 & 2 \\ -1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \left(\begin{array}{ccc|c} -2 & 1 & 2 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 1 & 0 & 0 \end{array}\right)$$

$$\sim\begin{pmatrix}-2 & 1 & 2 & 0\\ 0 & -1 & 2 & 0\\ -1 & 1 & 0 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-2 & 1 & 2 & 0\\ 0 & -1 & 2 & 0\\ 0 & 1 & -2 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-2 & 1 & 2 & 0\\ 0 & -1 & 2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-2 & 0 & 4 & 0\\ 0 & -1 & 2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix} \quad [R_1' = R_1 + R_2]$$

$$\sim\begin{pmatrix}1 & 0 & -2 & 0\\ 0 & 1 & -2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$$

$x_1 - 2x_3 = 0$

$x_2 - 2x_3 = 0$

$x_3 = t$ [free variable]

$x_2 = 2t$

$x_1 = 2t$

$$\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}2t\\ 2t\\ t\end{pmatrix} = t\begin{pmatrix}2\\ 2\\ 1\end{pmatrix}$$

$$V_3 = \begin{pmatrix}2\\ 2\\ 1\end{pmatrix}$$

For $\lambda=1$: $V_1 = \begin{pmatrix}0\\ -2\\ 1\end{pmatrix}$

For $\lambda=2$: $V_2 = \begin{pmatrix}1\\ 1\\ 0\end{pmatrix}$

For $\lambda=3$: $V_3 = \begin{pmatrix}2\\ 2\\ 1\end{pmatrix}$.

Q. Find the eigenvectors of A where 5, 3 and 3 are the eigenvalues of matrix A.

$$A = \begin{pmatrix}4 & 1 & -1\\ 2 & 5 & -2\\ 1 & 1 & 2\end{pmatrix}$$

Soln: For $\lambda=5$, let $\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}$ be the eigenvector

$$\begin{pmatrix}4-\lambda & 1 & -1\\ 2 & 5-\lambda & -2\\ 1 & 1 & 2-\lambda\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-1 & 1 & -1\\ 2 & 0 & -2\\ 1 & 1 & -3\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & -1 & 0\\ 2 & 0 & -2 & 0\\ 1 & 1 & -3 & 0\end{array}\right)$$

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$$\sim\begin{pmatrix}-1 & 1 & -1 & 0\\ 2 & 0 & -2 & 0\\ 1 & 1 & -3 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-1 & 1 & -1 & 0\\ 0 & 2 & -4 & 0\\ 0 & 2 & -4 & 0\end{pmatrix}\begin{matrix}[r_2'=r_2+2r_1]\\ [r_3'=r_3+r_1]\end{matrix}$$

$$\sim\begin{pmatrix}-1 & 1 & -1 & 0\\ 0 & 2 & -4 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-1 & 1 & -1 & 0\\ 0 & 1 & -2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}\left[r_2'=\frac{r_2}{2}\right]$$

$$\sim\begin{pmatrix}-1 & 0 & 1 & 0\\ 0 & 1 & -2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}\begin{matrix}[r_2'=r_2]\\ [r_1'=r_1-r_2]\end{matrix}$$

$$-x_1 + x_3 = 0$$
$$x_2 - 2x_3 = 0$$

$x_3 = t$

$$\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}t\\ 2t\\ t\end{pmatrix} = \begin{pmatrix}1\\ 2\\ 1\end{pmatrix}t$$

$$V_1 = \begin{pmatrix}1\\ 2\\ 1\end{pmatrix}$$

For $\lambda = 3$, let $\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}$ be the eigenvector.

$$\begin{pmatrix}4-\lambda & 1 & -1\\ 2 & 5-\lambda & -2\\ 1 & 1 & 2-\lambda\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\begin{pmatrix}1 & 1 & -1\\ 2 & 2 & -2\\ 1 & 1 & -1\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\begin{pmatrix}1 & 1 & -1 & 0\\ 2 & 2 & -2 & 0\\ 1 & 1 & -1 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}1 & 1 & -1 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$$

$$x_1 + x_2 - x_3 = 0$$

free $\rightarrow x_2 = t_1$

$x_3 = t_2$

$$\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix} = \begin{pmatrix}-t_1 + t_2\\ t_1\\ t_2\end{pmatrix} = \begin{pmatrix}-1\\ 1\\ 0\end{pmatrix}t_1 + \begin{pmatrix}1\\ 0\\ 1\end{pmatrix}t_2$$

$$V_2 = \begin{pmatrix}-1\\ 1\\ 0\end{pmatrix}, \quad V_3 = \begin{pmatrix}1\\ 0\\ 1\end{pmatrix}$$

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Q. Find the eigenvectors of A where $-3, -2$ and -2 are the eigenvalues of the matrix A.

$$A = \begin{pmatrix} 0 & -6 & -4 \\ 5 & -11 & -6 \\ -6 & 9 & 4 \end{pmatrix}$$

Soln: For $\lambda = -3$: Let, $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector

$$\begin{pmatrix} 0-\lambda & -6 & -4 \\ 5 & -11-\lambda & -6 \\ -6 & 9 & 4-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$3x_1 - 6x_2 - 4x_3 = 0$

$6x_2 + 2x_3 = 0$

$x_3 = t$ [free]

$x_2 = -\frac{1}{3}x_3 = -\frac{t}{3}$

$x_1 = \frac{6x_2 + 4x_3}{3}$

$$\sim \begin{pmatrix} 3 & -6 & -4 \\ 5 & -8 & -6 \\ -6 & 9 & 7 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 3 & -6 & -4 & 0 \\ 5 & -8 & -6 & 0 \\ -6 & 9 & 7 & 0 \end{array}\right)$$

$\Rightarrow x_1 = \frac{-2t + 4t}{3} = \frac{2t}{3}$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2t/3 \\ -t/3 \\ t \end{pmatrix} = t \begin{pmatrix} 2/3 \\ -1/3 \\ 1 \end{pmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 3 & -6 & -4 & 0 \\ 5 & -8 & -6 & 0 \\ -6 & 9 & 7 & 0 \end{array}\right)$$

$$= t/3 \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 3 & -6 & -4 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & -3 & -1 & 0 \end{array}\right)$$

$$\therefore V_1 = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

$$\sim \left(\begin{array}{ccc|c} 3 & -6 & -4 & 0 \\ 0 & 6 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

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For $\lambda=-2$: let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector:

$$\begin{pmatrix} 0-\lambda & -6 & -4 \\ 5 & -11-\lambda & -6 \\ -6 & 9 & 4-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim\begin{pmatrix} 2 & -6 & -4 \\ 5 & -9 & -6 \\ -6 & 9 & 6 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim\left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 5 & -9 & -6 & 0 \\ -6 & 9 & 6 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 5 & -9 & -6 & 0 \\ -6 & 9 & 6 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 0 & 12 & 8 & 0 \\ 0 & -9 & -6 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} 2 & -6 & -4 & 0 \\ 0 & 12 & 8 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$2x_1-6x_2-4x_3=0$

$12x_2+8x_3=0$

$x_3=t$ [free]

$x_2=-\frac{8}{12}t=-\frac{2}{3}t$

$x_1=\frac{6x_2+4x_3}{2}$

$=3x_2+2x_3$

$=3\left(-\frac{2}{3}t\right)+2t$

$=-2t+2t$

$\therefore x_1=0$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ -\frac{2}{3}t \\ t \end{pmatrix}=t\begin{pmatrix} 0 \\ -\frac{2}{3} \\ 1 \end{pmatrix}=\begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}\frac{t}{3}$$

$\therefore V_2=\begin{pmatrix} 0 \\ -2 \\ 3 \end{pmatrix}$.

* Complete Eigenspace:

$A_{n\times n}$ → n is eigen value

→ If we can find n independent eigenvalue

⇒ Complete eigenspace

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Diagonalization

$$P^{-1}AP = \text{diagonal}$$

Diagonal matrix $\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}, \begin{pmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 5 \end{pmatrix}$

$$P^{-1}AP = D$$

$$\Rightarrow \boxed{A = PDP^{-1}}$$

↳ Diagonal matrix

Diagonalization:

Let, A be an $n \times n$ matrix. If there exists an invertible $n \times n$ matrix P such that

$$P^{-1}AP = D \quad \text{or} \quad \boxed{A = PDP^{-1}}$$

where, D is an $n \times n$ diagonal matrix with eigenvalues on the diagonal.

Proof: $P^{-1}AP = D$

$\Rightarrow P^{-1}APP^{-1} = DP^{-1}$

$\Rightarrow P^{-1}AI = DP^{-1}$

$\Rightarrow P^{-1}A = DP^{-1}$

$\Rightarrow PP^{-1}A = PDP^{-1}$

$\Rightarrow IA = PDP^{-1}$

$\Rightarrow \boxed{A = PDP^{-1}}$

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Q: Show that A is Diagonalizable Matrix. Also find the basis of the eigenspace of A.

$$A=\begin{pmatrix}1 & 1 & 2\\ 0 & 2 & 2\\ -1 & 1 & 3\end{pmatrix}$$

Soln: Find λ: $|A-\lambda I|=0$

$$\begin{vmatrix}1-\lambda & 1 & 2\\ 0 & 2-\lambda & 2\\ -1 & 1 & 3-\lambda\end{vmatrix}=0$$

$\Rightarrow (1-\lambda)[(2-\lambda)(3-\lambda)-2]-1[0+2]+2[0+2-\lambda]=0$

$\Rightarrow (1-\lambda)(6-5\lambda+\lambda^2-2)-2+4-2\lambda=0$

$\Rightarrow 4-5\lambda+\lambda^2-4\lambda+5\lambda^2-\lambda^3+2-2\lambda=0$

$\Rightarrow \lambda^3-6\lambda^2+11\lambda-6=0$

$\therefore \lambda=1,2,3.$

For $\lambda=1$: Let $\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}$ be the eigenvector

$$\begin{pmatrix}1-\lambda & 1 & 2\\ 0 & 2-\lambda & 2\\ -1 & 1 & 3-\lambda\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\begin{pmatrix}0 & 1 & 2\\ 0 & 1 & 2\\ -1 & 1 & 2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}$$

$$\sim\left(\begin{array}{ccc|c}0 & 1 & 2 & 0\\ 0 & 1 & 2 & 0\\ -1 & 1 & 2 & 0\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & 2 & 0\\ 0 & 1 & 2 & 0\\ 0 & 1 & 2 & 0\end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c}-1 & 1 & 2 & 0\\ 0 & 1 & 2 & 0\\ 0 & 0 & 0 & 0\end{array}\right)$$

$$\sim\begin{pmatrix}-1 & 0 & 0\\ 0 & 1 & 2\\ 0 & 0 & 0\end{pmatrix}\begin{pmatrix}0\\ 0\\ 0\end{pmatrix}=0$$

$-x_1=0$

$x_2+2x_3=0$

$x_3=t$ [free]

$$\begin{pmatrix}x_1\\ x_2\\ x_3\end{pmatrix}=\begin{pmatrix}0\\ -2t\\ t\end{pmatrix}=\begin{pmatrix}0\\ -2\\ 1\end{pmatrix}t$$

$$V_1=\begin{pmatrix}0\\ -2\\ 1\end{pmatrix}$$

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For $\lambda=2$, let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector

$$\begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim\begin{pmatrix} -1 & 1 & 2 \\ 0 & 0 & 2 \\ -1 & 1 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$-x_1+x_2=0$

$2x_3=0$

$x_2=t$ [free]

$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} t \\ t \\ 0 \end{pmatrix}=\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}t$

$$\sim\left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ -1 & 1 & 1 & 0 \end{array}\right)$$

$V_2=\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

$$\sim\left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} -1 & 1 & 2 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

$$\sim\left(\begin{array}{ccc|c} -1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

For $\lambda=3$: let $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ be the eigenvector

$$\begin{pmatrix} 1-\lambda & 1 & 2 \\ 0 & 2-\lambda & 2 \\ -1 & 1 & 3-\lambda \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim\begin{pmatrix} -2 & 1 & 2 \\ 0 & -1 & 2 \\ -1 & 1 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\sim\left(\begin{array}{ccc|c} -2 & 1 & 2 & 0 \\ 0 & -1 & 2 & 0 \\ -1 & 1 & 0 & 0 \end{array}\right)$$

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$$\sim\begin{pmatrix}-2 & -1 & 0 & 0\\ 0 & -1 & 2 & 0\\ -1 & 1 & 0 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-2 & -1 & 2 & 0\\ 0 & -1 & 2 & 0\\ 0 & 1 & -2 & 0\end{pmatrix}$$

$$\sim\begin{pmatrix}-2 & -1 & 2 & 0\\ 0 & -1 & 2 & 0\\ 0 & 0 & 0 & 0\end{pmatrix}$$

$V_3 = \begin{pmatrix}2\\2\\1\end{pmatrix}$

$\lambda=1$	$\lambda=2$	$\lambda=3$
$V_1=\begin{pmatrix}0\\-2\\1\end{pmatrix}$	$V_2=\begin{pmatrix}1\\1\\0\end{pmatrix}$	$V_3=\begin{pmatrix}2\\2\\1\end{pmatrix}$

$$D=\begin{pmatrix}2 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 3\end{pmatrix} \rightarrow P=\begin{pmatrix}1 & 0 & 2\\ 1 & -2 & 2\\ 0 & 1 & 1\end{pmatrix}$$

$$\text{OR, } D=\begin{pmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ 0 & 0 & 3\end{pmatrix} \rightarrow P=\begin{pmatrix}0 & 1 & 2\\ -2 & 1 & 2\\ 1 & 0 & 1\end{pmatrix}$$

$$P^{-1}=\begin{pmatrix}\frac{1}{2} & -\frac{1}{2} & 0\\ 2 & -1 & -2\\ -\frac{1}{2} & \frac{1}{2} & 1\end{pmatrix}$$

$P^{-1}AP = D$

$A = PDP^{-1}$

Verify.

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